Experiment - 8

In [1]:

```
import cv2 import
numpy as np

# Step 1: Open video
cap = cv2.VideoCapture("videoo.mp4")

# Step 2: Read first frame
ret, old_frame = cap.read()
old_gray = cv2.cvtColor(old_frame, cv2.COLOR_BGR2GRAY)

# Step 3: Detect feature points using Shi-Tomasi p0
= cv2.goodFeaturesToTrack(old_gray, 100, 0.3, 7)
while True:

    # Step 4: Read next frame     ret, frame =
cap.read()     if not ret:         break     gray =
cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    # Step 5: Apply Lucas-Kanade Optical Flow
p1, status, _ = cv2.calcOpticalFlowPyrLK(
old_gray, gray, p0, None     )

    # Step 6: Draw motion trajectories     for new, old in
zip(p1[status == 1], p0[status == 1]):
        x1, y1 = new.ravel()
x2, y2 = old.ravel()

        cv2.arrowedLine(
frame,
            (int(x2), int(y2)),
            (int(x1), int(y1)),
            (0, 255, 0), 2
)

        # Calculate motion
dx = x1 - x2         dy =
y1 - y2
        speed = np.sqrt(dx**2 + dy**2)

    # Step 7: Display tracking result
cv2.putText(frame, "Object Tracking",
                (20, 40), cv2.FONT_HERSHEY_SIMPLEX,
                1, (0, 255, 255), 2)

cv2.imshow("Optical Flow Tracking", frame)

    # Step 8: Update previous frame and points
old_gray = gray
    p0 = p1[status == 1].reshape(-1, 1, 2)

    # Step 9: Press Q to exit
```

```
    if cv2.waitKey(30) & 0xFF == ord('q'):
break

# Step 10: Release resources
cap.release()
cv2.destroyAllWindows()
```

